



Quiet by Other Means: Interviewing Grounds Staff About a Courtyard Deterrent's Compliance Log

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Abstract. Grounds and facilities staff at Slop University's central courtyard offset an automated bioacoustic deterrent installed to hold the pigeon population below a nuisance threshold. We report a semi-structured interview study ($n = 11$) with staff who maintain the courtyard and the buildings around it, alongside the deterrent's own compliance log across the same eleven-week window. Staff described a set of undocumented routines — closing specific windows before dusk, retiming bin collection around the courtyard's quiet hours, and quietly tolerating the device's occasional fault during exam weeks rather than escalating for repair — that they understood as the mechanism actually holding sightings down, distinct from and largely unremarked by the device's own reporting. A weekly Routine Coverage Index relating staff-reported interventions to the device's logged clear-day count held in a narrow band across the window (mean 0.48), consistent with the two series moving together rather than either leading. A nine-day scheduled outage let us compare sightings with the device silent: the mean held steady (2.18 sightings per day against 2.06 with the device active), a difference well within the spread of either condition, while staff routines continued unchanged throughout. We read the correspondence as an instance of formal instrumentation defaulting to full credit for outcomes an undocumented routine may be doing some or all of the work to produce, and discuss the consequences for how the University attributes environmental outcomes to sensed infrastructure rather than to the staff who tend it.

Introduction

An automated bioacoustic deterrent went into service in the University's central courtyard three terms ago, installed after repeated pigeon-related maintenance tickets pushed the courtyard onto the estate team's watch list. The device broadcasts a rotating library of distress and predator calls on a schedule set by its own sensor readings, and reports its performance monthly as a single figure: the share of days logged clear of a sustained roosting event. That figure has improved steadily since installation, and the estate team's monthly report attributes the whole of the improvement to the device.

Grounds and facilities staff who work in and around the courtyard describe a different picture in passing conversation: windows closed earlier than the roster requires, bin collection re-

timed around the courtyard's quiet hours, and a tacit understanding not to escalate the device's own faults during exam weeks, when taking the courtyard offline for repair is judged worse than tolerating an occasional false negative. None of this appears in any log the estate team reviews. We treat the gap between what staff do and what the compliance report credits as an empirical question rather than an aside, and ask what a systematic account of the undocumented routine adds to the device's own record of its performance.

We report a semi-structured interview study with eleven staff who maintain the courtyard or the buildings around it, paired with the deterrent's compliance log over the same eleven-week window. The study makes three contributions, each hedged to the evidence it can bear:

- We provide, to our knowledge, the first systematic account of the informal routines Slop University grounds staff maintain around an automated wildlife-deterrent installation, though the account is necessarily local to one courtyard and one team.
- We show that a weekly index relating staff-reported interventions to the device's own clear-day count holds in a narrow, stable band across the fieldwork window, suggesting the two series are not independent, though the study cannot establish which, if either, is doing the work.
- We report a natural comparison across the device's nine-day scheduled outage in which staff routines continued unchanged; sightings during the outage were statistically indistinguishable from sightings with the device active, a result we read cautiously given the outage's short duration.

The remainder of the paper situates the study against work on invisible and articulation labour, on the reactivity of institutional metrics, and on bird-deterrence technology; describes the interview and log-linking method; reports the routine-log correspondence and the outage comparison; and discusses what the University's attribution practice might be missing.

Related work

The gap between the labour a system is credited with and the labour that actually sustains it is a long-standing concern in studies of work. Star and Strauss's account of the "ecology of visible and invisible work" names the pattern directly: work that is routine, distributed, or deliberately kept off the record is systematically undercounted by the formal apparatus built to track performance [1]. Hampson and Junor extend the point to service settings, where staff routinely perform "articulation work" that holds a formal process together without appearing anywhere the process is measured [2]. Studies of electronic health records describe closely related workarounds: staff maintain informal talk and manual routines around systems that are reported, upward, as functioning as designed [3], [4].

A parallel literature treats the metric itself as an actor rather than a passive record. Strathern's collection on audit culture argues that the instruments an institution uses to hold itself

accountable reshape the practice they measure [5]; Espeland and Sauder show that public rankings and similar measures are reactive, altering the behaviour they claim only to describe [6], and Espeland and Stevens trace commensuration — the reduction of qualitatively different practices to a single comparable figure — as a general social process with this same reshaping effect [7]. Muller's account of "metric fixation" and the Goodhart's-law literature on gamed indicators describe the resulting failure mode, in which a measure substitutes for the thing it was meant to track [8], [9], [10]. Evans situates the discretion street-level staff exercise around formal rules in the same frame [11]. On the deterrent technology itself, acoustic and visual bird-scaring devices have a documented but modest evidence base for sustained effectiveness [12], [13], [14], [15]. Mathematical models of urban pigeon populations suggest management strategies interact with population dynamics in ways a single compliance figure cannot capture [16], and bioacoustic classification work shows that distinguishing a genuine target signal from ordinary human and bird activity in a field recording is itself a nontrivial discrimination problem [17]. Goodhart's-law effects have been documented well outside institutional research settings — corrective vehicle-emissions policy among them [18] — which motivated our interest in what else might be doing the work here. The deterrent itself belongs to a fast-moving category of eco-acoustic systems still being refined in the engineering literature [19].

Method

Site and recruitment. The study site is the courtyard connecting two teaching buildings, home to the automated bioacoustic deterrent since its installation three terms prior. We recruited eleven staff (six grounds staff, three facilities technicians, two building attendants) via the estate team's roster, purposively sampling for at least one full working week of exposure to the courtyard within the fieldwork window. Interviews ran forty to fifty-five minutes, were audio-recorded with consent, and were held off the estate team's premises to reduce any perceived link to the compliance report.

Interview protocol. Semi-structured interviews opened with a walkthrough of a typical shift, moved to specific probes about bin timing, window closure, and refuse collection near the

courtyard, and closed with a direct question about whether the participant believed the device's monthly figure reflected what they did. We coded transcripts with an open-coding pass followed by axial grouping into routine categories (window closure, bin retiming, refuse delay, informal fault tolerance), with saturation — no new routine category emerging across three consecutive interviews — reached at interview nine.

Log linkage. The deterrent logs a binary daily reading (clear / roosting event) from its own sensor pass, aggregated by the estate team into a weekly clear-day count. We matched each fieldwork week's interview-derived count of distinct staff-reported interventions against that week's logged clear-day count, and defined a weekly Routine Coverage Index:

$$RCI_w = \frac{S_w}{S_w + L_w}$$

where S_w is the number of staff-reported interventions in week w and L_w is the device's own clear-day count for the same week. An index near 0.5 indicates the two series move together; values drifting toward 1 would suggest staff activity outpacing what the log credits, and values drifting toward 0 the reverse. We report the index alongside the raw series rather than as a single summary figure, since its behaviour across the window is itself part of the finding.

Outage comparison. A nine-day scheduled maintenance outage fell within the fieldwork window. We treated it as a naturally occurring ablation: the device off, staff routines — by every account — unchanged, and compared daily estate-logged pigeon sightings across the two conditions with a two-sample comparison of means.

Results

Staff-reported interventions track the device's own record. Figure 1 plots the weekly count of staff-reported interventions against the device's logged clear-day count across the eleven-week window. The two series rise together ($r = 0.90$); weeks in which staff described tightening their bin and window routine are, without exception, weeks in which the device's own log also improved.

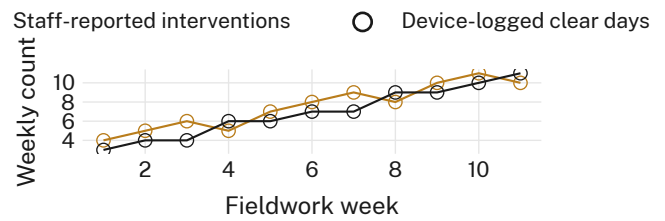


Figure 1: Weekly staff-reported routine interventions and the deterrent's own logged clear-day count rise together across the fieldwork window (r is approximately 0.90); the estate team's monthly report attributes the rise entirely to the device.

The Routine Coverage Index held in a narrow band across the window (mean 0.48, range 0.41-0.55), consistent with the two series moving together rather than either one leading; the index shows no drift toward either extreme across the eleven weeks, so the fieldwork window cannot distinguish which of the two is, in a causal sense, doing the work.

Role	n	Median weekly courtyard hours
Grounds staff	6	11
Facilities technicians	3	4
Building attendants	2	7

The outage changed nothing the log would notice. The scheduled nine-day outage let us compare sightings with the device silent against sightings with it active. Figure 2 shows near-identical distributions: a mean of 2.06 sightings per day with the device active against 2.18 during the outage, a difference well within the spread of either condition and, if anything, in the direction opposite to what the device's own attribution would predict.

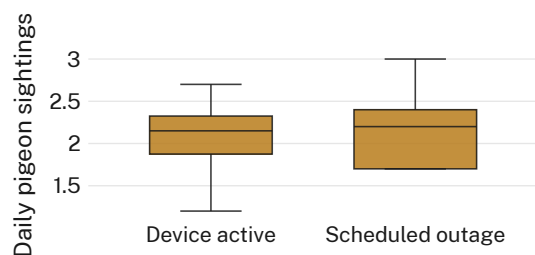


Figure 2: Daily pigeon sightings during the deterrent's nine-day scheduled outage (mean 2.18/day) were statistically indistinguishable from sightings with the device active (mean 2.06/day); we retain the device in service pending a longer trial.

Nine of eleven participants independently raised the same explanation when asked why sightings held steady during the outage: the window-closure and bin-retiming routine ran on the cleaning and refuse rosters, not on the device's schedule, and neither roster paused for the outage. Two participants noted, unprompted, that they had not mentioned the outage to the estate team at the time, judging the distinction between "device active" and "device offline" immaterial to their own routine.

Discussion

The correspondence in Figure 1, and the outage comparison in Figure 2, are each consistent with a simple explanation: the routines staff described are doing observable work, and the device's monthly figure is crediting itself with a share of that work by default, in the absence of any competing record. We do not claim the device contributes nothing — our design cannot separate its contribution from the routine's — only that the estate team's current attribution practice has no mechanism for finding out. A compliance report that can only see what it logs will, unsurprisingly, credit itself with everything within its field of view.

The pattern sits alongside a wider finding in the invisible-work literature: labour that is distributed, routine, and never entered into a system of record tends to disappear from that system's account of its own success, not through any particular actor's decision but through the ordinary operation of a report that samples only what it was built to sample. Extending the fieldwork window past a single scheduled outage, and instrumenting the routine itself rather than relying on interview recall, would let a future study

separate the two contributions the present design can only show move together.

Limitations

Our account rests on eleven interviews and one courtyard, and a single nine-day outage is a narrow window on which to base a causal claim; we report the outage comparison descriptively rather than as a controlled trial. Interview recall of routine timing may itself be imprecise, though the convergence across nine of eleven participants on the same explanation for the outage result suggests the pattern is not an artefact of any one account. The Routine Coverage Index is a ratio of two counts collected by different means (interview recall and device sensor) and inherits the imprecision of both; we present it descriptively rather than as a validated instrument. A larger, multi-site study would let the two coverage mechanisms be told apart with more confidence than the present design allows.

Conclusion

Grounds and facilities staff maintaining Slop University's central courtyard sustain a set of undocumented routines that track closely with the compliance figure an automated bioacoustic deterrent reports for itself, and sightings during the device's own nine-day outage were no worse than sightings with it running. We read the correspondence as evidence that the University's current attribution practice has no way of crediting the routine separately from the device, not as evidence that the device is unnecessary. Future work should instrument the routine directly rather than relying on interview recall, and extend the comparison across a longer or repeated outage. Until then, the courtyard's compliance report and the staff who keep it quiet are, on the present evidence, describing the same result from two very different vantage points.

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